

Preregistered Paired Evaluation of a Deterministic Verifier Pipeline for LLM-Generated Network Configuration Changes — Non-informative Result under Shared-Generation Semantics

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Abstract—We report a fully preregistered paired experiment (cand-2026-01-prereg-v1) that tested whether a deterministic verifier pipeline (generate → deterministic_netops_validator_v5 → optional repair) reduces the rate of verifier-detected unsafe intent→configuration proposals produced by a hosted LLM on synthetic NetOps tasks. Design: N=200 preregistered unique intent→configuration tasks in a paired design comparing (a) baseline LLM-only and (b) guarded pipeline (gpt-5-mini + deterministic_netops_validator_v5). Execution used shared_initial_candidate generation semantics (one LLM call per task) producing model_calls_used = 200 (preregistered independent per-arm generation would have used up to 400). Observed (adversarial cluster): baseline SAFE = 200/200; guarded SAFE = 200/200 (paired contingency n11 = 200, n10 = 0, n01 = 0). Estimated paired SAFE-rate difference = 0.0; exact McNemar two-sided $p = 1.0$; paired bootstrap 95% CI = [0.0, 0.0] (resamples = 10000, seed = 1729). Because identical per-pair generations were produced and the observed baseline unsafe rate was 0%, the preregistered causal hypothesis (that the deterministic verifier reduces unsafe proposals) was not testable in this execution. We publish the complete preregistered run and artifacts, provide an artifact-grounded diagnosis of testability failure, and give concrete, archive-supported recommendations (independent per-arm generation budgeting, adversarial injection calibration, and exploratory ROC reserves) for informative repeats. All execution and analysis artifacts cited here are archived in the supplied bundle.

I. INTRODUCTION

Automating human-intent → device-configuration translation with large language models (LLMs) is an active NetOps research thrust because it can reduce human effort and speed maintenance tasks. Yet incorrectly specified or misordered changes can cause outages, policy violations, or security exposures. A commonly proposed mitigation is tool-augmentation: couple an LLM generator to deterministic validators, sandboxed simulators, or constrained executors in a generate-validate-repair pipeline so that proposed changes are checked before execution.

We preregistered and executed a confirmatory paired experiment (cand-2026-01-prereg-v1) to quantify whether deterministic verifier augmentation causally reduces the frequency of verifier-detected unsafe proposals from a hosted LLM under both benign and adversarial prompt clusters. The preregistered estimand was the paired SAFE-rate difference (guarded minus

baseline), tested by an exact McNemar two-sided test with a paired bootstrap CI (10000 resamples, seed = 1729). The design sampled N=200 tasks using a deterministic generator and a paired comparison across two pipelines. Two generation semantics were permitted in the preregistration: independent per-arm generation (one LLM call per arm per task) or shared_initial_candidate (one shared LLM call per task scored by both arms). The executed run used shared_initial_candidate to conserve model-call budget.

Key outcome: under the executed shared semantics the sampled LLM outputs were scored SAFE by the deterministic validator in every confirmatory episode (baseline SAFE = 200/200; guarded SAFE = 200/200). This concordant ceiling outcome (n11 = 200; n10 = n01 = n00 = 0) yields an estimate of zero paired difference and an exact McNemar $p = 1.0$. Because the observed baseline unsafe rate was 0% and the same candidate was used in both arms, the preregistered causal hypothesis could not be meaningfully evaluated. The manuscript therefore focuses on three contributions grounded in archived artifacts: (1) a complete, deterministic preregistered run published as reproducible artifacts; (2) a precise, artifact-grounded diagnosis of why this execution was non-informative for the confirmatory estimand; and (3) concrete, artifact-supported design recommendations that would enable informative repeats under the same preregistration framework.

II. RELATED WORK

Recent surveys and prototypes explore LLMs for NetOps tasks including intent translation, configuration generation, and remediation planning; these works emphasize both potential productivity gains and safety concerns when LLMs propose executable changes [openalex:W7161354925; openalex:W7170714602; TechRxiv:177004277]. Prior empirical and architectural work shows that augmenting LLMs with deterministic tools (verifiers, simulators, guarded tool schemas) often improves measured correctness on curated task sets and can reduce constraint violations in produce-validate pipelines [openalex:W4412888330; openalex:W7161354925]. Red-teaming, guardrail, and MCP/schema proposals highlight trade-offs between missed unsafe proposals and increased false refusals, and they call for workflow-centred, replayable

evaluations with deterministic sandboxing and archived traces [TechRxiv:177006109; TechRxiv:177004277].

Our study is not a new verification algorithm or a multi-model leaderboard. Instead, it complements existing work by empirically interrogating how concrete experimental semantics—particularly generation budgeting and shared vs independent generation—interact with a paired estimand and can render a confirmatory design untestable. We therefore situate our contribution as a reproducible, methodological diagnosis that bridges architecture proposals and practical evaluation design.

III. METHODOLOGY

Preregistration and primary estimand. The experiment was preregistered as cand-2026-01-prereg-v1. The primary estimand is the paired binary difference `paired_success_rate_difference_guarded_minus_baseline` where each arm contributes a binary indicator per task (1 = SAFE, 0 = UNSAFE) derived from a deterministic sandbox validator. The confirmatory test is an exact two-sided McNemar implemented by the `paired_binary_analysis_v1` executor; paired bootstrap percentile confidence intervals use 10,000 resamples with seed = 1729. The preregistration specified a paired binary design with N=200 unique tasks (400 paired episodes across two arms under independent per-arm draws), power calculations targeting an ≈ 7 percentage-point paired reduction, and predeclared sensitivity and multiplicity procedures.

Task generation, scope, and deterministic seeding. A deterministic task generator (`deterministic_netops_task_generator_v5`) produced 200 synthetic intent→configuration tasks constrained to allowed fields {admin, mtu, vlan} within single-AS multi-device topologies. Each task manifest includes `initial_state`, `expected_final_state`, an explicit `required_sequence` (workflow policy), and difficulty metadata; generation was seeded deterministically (per-task `generation_seed` 1..200 as recorded in `execution/task_manifest.jsonl`) so every task can be reconstructed exactly. Tasks were classified into benign and adversarial clusters using a preregistered adversarial template set designed to increase difficulty by obfuscation, contradictory constraints, and privilege-escalation phrasing; the adversarial cluster formed the confirmatory challenge set alongside benign items per the preregistration.

Pipelines compared and validator semantics. Two pipelines were contrasted: (1) baseline (LLM-only): accept the generator output and evaluate post hoc with `deterministic_netops_validator_v5`; (2) guarded (LLM + deterministic validator): validate the generator output using `deterministic_netops_validator_v5` in a sandboxed replay and (optionally) invoke an automatic repair step once. For confirmatory episodes in this run the repair pathway was not exercised (`repair_provider_trace` fields are null in `execution/scoring/*.json`), so the guarded arm operationally reflects deterministic validation only. The validator applies the preregistered `scoring_rule_full_intent_constraint_validation` and emits deterministic traces

(`validation_before`, `validation_after`, violation lists, and a boolean valid flag) that produce the binary SAFE/UNSAFE outcome used by analysis. Representative validator traces are archived under `execution/scoring/*` and `execution/responses/*` and show the `validator_id` and `validator_version` recorded per episode.

Generation semantics, model budgeting, and execution accounting. The preregistration allowed two generation semantics: independent per-arm generation (two independent LLM calls per task, one for each arm) or `shared_initial_candidate` (one shared LLM call per task whose candidate is scored by both arms). To conserve hosted model budget the executed run used `shared_initial_candidate` semantics; this is recorded in the archived model configuration (`execution/model_configuration.json`: `independent_condition_generation` = false) and in the execution manifest (`execution/execution_manifest.json`: `model_calls_used` = 200). The preregistered independent per-arm design would have used up to 400 model calls (two per task) under the same N=200 confirmatory sample; that budget differential is therefore the precise additional model-call requirement to enable arm-level stochastic contrasts under the preregistered plan. The execution manifest and model configuration are archived and document these accounting facts.

Scoring, missingness, exclusions, and deterministic reconciliation. `Deterministic_netops_validator_v5` produced a deterministic valid flag per candidate; confirmatory binary labels were derived directly from `validator.valid`. The preregistration specified one automatic retry for model call failures, duplicate-prompt detection (prompt hashes lead to automatic exclusion+replacement), verifier nondeterminism checks (replays), and a stopping rule that aborts or downgrades if technical failures exceed 10%. In the archived confirmatory run, deterministic reconciliation artifacts (`analysis/deterministic_reconciliation.json` and `execution/execution_manifest.json`) report `completed_episode_count` = 400, `failed_episode_count` = 0, `complete_pairs` = 200, and that all deterministic consistency checks passed; `analysis/contamination_summary.csv` and `analysis/missingness_summary.csv` document zero exclusions or technical failures. Those artifacts confirm that the run executed under the shared generation semantics without technical interruptions and that no contaminated or nondeterministic tasks were present in the confirmatory set.

Analysis pipeline and confirmatory computation. The analysis executor `paired_binary_analysis_v1` consumed the per-episode binary outcomes and produced the 2×2 contingency (`analysis/paired_contingency_table.csv`: n11,n10,n01,n00), exact McNemar two-sided p-value, and paired bootstrap CI (`analysis/analysis_log.jsonl`). For this execution the contingency was n11 = 200, n10 = 0, n01 = 0, n00 = 0, yielding an estimated paired SAFE-rate difference = 0.0, exact McNemar p = 1.0, and paired bootstrap 95% CI = [0.0, 0.0]. Representative per-task

scoring JSONs and shared initial responses are archived under `execution/scoring/` and `execution/responses/` (e.g., `task-000001-*.json` and `task-000001-shared-initial.txt`) and show identical `shared_initial_candidate` content and validator traces with `violation_count = 0`. The preregistered exploratory reserve (up to 50 tasks to estimate verifier ROC/refusal trade-offs) was not consumed prior to confirmatory execution; consequently no ROC or refusal estimates appear in the confirmatory artifacts.

Design consequences and protocol fidelity. Because the executed semantics produced a single shared candidate per task that both arms scored deterministically and because the validator labeled every sampled candidate SAFE, the paired design produced no discordant pairs and therefore no testable paired effect under McNemar logic. This outcome is an execution-semantic and budgeting consequence that follows directly from archived configuration and execution artifacts (`execution/model_configuration.json` and `execution/execution_manifest.json`) rather than from properties of the validator algorithm. All files referenced above are included in the archived bundle so readers can replay generation, scoring, and analysis deterministically to confirm the chain of events described here.

IV. RESULTS

Execution accounting and provider audit. The execution manifest reports `completed_episode_count = 400` (200 paired episodes), `failed_episode_count = 0`, and `model_calls_used = 200`, consistent with the executed `shared_initial_candidate` semantics. Provider-level auditing recorded `hosted_model_call_event_count = 200` and `cache_miss_count = 200`; cross-condition cache reuse was not observed. No technical failures, exclusions, or nondeterministic verifier behaviors were recorded in the confirmatory set, and deterministic reconciliation checks passed, confirming that the archived episode and scoring artifacts correspond to a single, reproducible run.

Primary confirmatory outcomes and contingency. The deterministic validator produced SAFE labels for every confirmatory episode: baseline SAFE successes = 200/200 and guarded SAFE successes = 200/200. The resulting paired contingency counts are $n_{11} = 200$ (SAFE in both arms), $n_{10} = 0$ (SAFE guarded only), $n_{01} = 0$ (SAFE baseline only), and $n_{00} = 0$ (UNSAFE in both). From these counts the paired SAFE-rate difference (guarded - baseline) equals 0.0. The preregistered exact McNemar two-sided test computed $p = 1.0$; the paired bootstrap percentile 95% confidence interval is degenerate at $[0.0, 0.0]$ (resamples = 10000; bootstrap_seed = 1729). These numerical values are direct outputs of `paired_binary_analysis_v1` and are archived in the analysis artifacts (`condition_summary` and `paired_contingency_table` files).

Representative diagnostic traces. Representative scoring JSONs and validator traces illustrate why the contingency is degenerate. For multiple sampled tasks the

`shared_initial_candidate` content (the raw generated configuration) is identical across baseline and guarded episodes; validator traces show `validation_before` and `validation_after` with `violation_count = 0` and `valid = true` in every case. The `repair_provider_trace` entries are null in all confirmatory scoring JSONs, confirming that no automatic repair was invoked and that the guarded arm performed only validation. Shared response files for representative tasks contain the canonical generated configurations used by both arms, reinforcing that no arm-specific generation stochasticity existed under the executed semantics.

Missingness, contamination, and exploratory reserves. `Analysis/missingness_summary` documents `complete_pairs = 200` with zero failed or unscorable episodes; `analysis/contamination_summary` shows no flagged episodes. The preregistration reserved up to 50 exploratory tasks to estimate verifier ROC and refusal trade-offs; that exploratory reserve was not consumed prior to confirmatory execution, so no empirical ROC or refusal-rate estimates appear in the confirmatory analysis artifacts.

Statistical interpretation and inferential consequence. The observed ceiling outcome ($n_{11} = 200$ with no discordant pairs) renders the preregistered inferential machinery non-informative for the intended causal estimand: McNemar’s test relies on discordant pairs (n_{10} and n_{01}) to evaluate paired differences, and their absence produces a p -value of 1.0 and a degenerate CI at zero. This degeneracy does not demonstrate verifier effectiveness; rather, it indicates that under the executed shared generation semantics and for the sampled model outputs, there was no arm-level variation to attribute to the presence or absence of the deterministic validator. The preregistered power calculation had assumed a nonzero baseline unsafe rate ($\approx 10\text{--}15\%$) and discordant proportions; those assumptions were not met in the executed run because budgeted generation semantics constrained per-arm variability.

Archival fidelity and reproducibility pointers. All numerical summaries and the contingency table reported above are identical to the archived analysis artifacts (`analysis/condition_summary.csv` and `analysis/paired_contingency_table.csv`). Representative per-task scoring JSONs and shared response files show the identical generated candidate for baseline and guarded episodes and document validator decisions and violation lists. The execution and analysis logs capture the exact bootstrap parameters (resamples = 10000, seed = 1729) and the exact statistical outputs. Because the confirmatory run contains no technical failures or exclusions, the archived artifacts permit deterministic replay of the identical non-informative outcome; investigators seeking an informative repeat should consult these artifacts before changing preregistered design choices.

V. DISCUSSION

We expand the diagnostic interpretation and practical implications of the observed non-informative (ceiling) outcome, grounding each statement in archived execution and analysis artifacts.

Root cause synthesis. The degenerate contingency ($n_{11}=200$, $n_{10}=n_{01}=n_{00}=0$) results from the conjunction of three archived facts: (1) generation semantics: the run executed `shared_initial_candidate` (`independent_condition_generation = false`) and therefore produced exactly one LLM candidate per task that both arms scored; (2) validator outcomes: `deterministic_netops_validator_v5` labeled every sampled candidate as valid (baseline SAFE = 200/200); and (3) absence of repairs: `repair_provider_trace` fields are null in confirmatory scoring JSONs, so the guarded arm did not alter any candidate. These archived facts (`execution/model_configuration.json`, `execution/execution_manifest.json`, `execution/scoring/*.json`, `analysis/paired_contingency_table.csv`) together make the paired causal contrast mechanically untestable because the arms had no opportunity to disagree.

Implications for confirmatory design. A paired estimand that attributes differences to the presence of a deterministic verifier requires per-arm variation that could produce discordant outcomes. Under shared generation semantics, discordance can only arise from repair-driven changes in the guarded arm or nondeterministic validator behavior; neither occurred here. Accordingly, the preregistered power calculation—which assumed a baseline unsafe rate $\approx 10\text{--}15\%$ and discordant proportions that yield detectability with $N \approx 174$ —no longer applied. The archived `model_call` accounting quantifies the specific resource trade-off: independent per-arm generation would have required ≈ 400 model calls (preregistered `maximum_model_calls`), whereas the executed shared semantics used 200 calls (`execution/execution_manifest.json`). When budgets force a choice, the estimand should drive budgeting: per-arm independence is necessary to test the causal effect of adding a validator to generation.

Practical recommendations for informative repeats (artifact-grounded). All recommendations follow directly from the preregistration and execution artifacts: - Use independent per-arm generation for confirmatory paired contrasts unless the guarded arm is capable of deterministic repairs that will reliably produce candidate changes. For this study size ($N=200$), independent generation requires roughly 200 additional model calls (400 total) compared with the executed shared run (200 `model_calls_used`). This arithmetic is visible in the preregistered `execution_contract` and the executed `manifest`. - Prior to running the confirmatory set, consume a preregistered exploratory reserve (we reserved up to 50 tasks) to calibrate adversarial transforms until the baseline unsafe rate matches the power assumptions. Archive these tuning runs and fix the transform parameters before the confirmatory execution so the confirmatory assumptions remain preregistered and reproducible. - If repairs are part of the guarded pipeline, instrument, archive, and analyze `repair_provider_trace` artifacts: quantify the fraction of UNSAFE→SAFE conversions and check whether repairs introduce new violations. In our run repair traces were null; therefore repairs did not serve as a mechanism to induce discordance. - Implement a deterministic pilot check step that inspects the contingency counts on a small holdout before

committing to the full confirmatory test. If the pilot produces a ceiling/floor contingency (no discordant pairs), abort and adapt along preregistered contingency-preserving paths (e.g., enable independent per-arm draws or strengthen adversarial transforms) rather than proceeding to a confirmatory test that will be uninformative.

Alternative estimands and complementary analyses. When budgets or policy preclude independent per-arm draws, consider changing the estimand to one that remains testable under shared semantics: (a) estimate the absolute unsafe-proposal rate of the LLM baseline (single-arm proportion) with appropriate CIs; (b) treat the guarded pipeline as an accept/reject gate and estimate refusal and post-repair SAFE rates (requires repairs and repair traces); or (c) treat the validator as a diagnostic tool and run exploratory ROC estimation (requires ground truth UNSAFE injections and the exploratory reserve). These alternative estimands shift the required resources but preserve inferential validity; the preregistration explicitly listed several `secondary_estimands` and exploratory items that map to these alternatives.

Threats to generalization and validator completeness. The deterministic validator used here encodes a formalized `full_intent_constraint_validation` for the synthetic task family and ran deterministically in the sandboxed environment; however, its completeness relative to production failure modes is limited. The synthetic tasks restrict fields to `admin/mtu/vlan` and single-AS topologies; consequently, zero unsafe detections here do not imply general safety in production multi-vendor or multi-AS contexts. Future work should validate validators against broader operational rule sets and consider human adjudication for ambiguous cases; such extensions must be preregistered to avoid post-hoc instrument selection.

Operational implications. From an operator perspective, strict guardrails can reduce risk but may also increase false refusals and operator workload. Our archived run did not produce refusals, so we cannot quantify that trade-off here; nevertheless, the preregistered exploratory ROC reserve was intended to measure precisely this tension and should be consumed in future repetitions. For deployment decisions, teams should balance the cost of additional model calls (for independent draws or exploratory calibration) against the operational risk of executing unverified changes; our artifacts make the per-task resource implications explicit and replayable.

Reproducibility and reuse. All artifacts supporting the above diagnosis and recommendations are archived (`execution/*` and `analysis/*`). They permit deterministic replay of the executed semantics (`shared_initial_candidate`), verification of `model_call` accounting (`model_calls_used = 200` vs planned 400), and inspection of representative scoring traces that expose why no arm-level discordance occurred. We encourage future repeats to adopt the pilot-check and exploratory calibration steps described above and to record the exact decision points in the preregistration so that execution semantics remain auditable and the confirmatory estimand remains testable.

VI. LIMITATIONS

- Synthetic tasks and scope: tasks were deterministic synthetic topologies produced by `deterministic_netops_task_generator_v5` and limited to single-AS, multi-device setups; results may not generalize to production, multi-AS, or vendor-specific features.
- Generation semantics: the executed run used `shared_initial_candidate` (`independent_condition_generation = false`), producing identical per-pair generations and creating a ceiling effect when shared candidates were valid.
- Adversarial strength: the preregistered adversarial templates used in this run did not elicit unsafe outputs from `gpt-5-mini` on the sampled tasks; stronger adversarial cases or explicit injected unsafe examples are required to measure verifier effect.
- Single-model scope: only the declared model (`gpt-5-mini`) was evaluated; no multi-model generality is claimed.
- Verifier completeness: `deterministic_netops_validator_v5` ran deterministically in synthetic sandboxed states; external validation of validator completeness against all real-world NetOps failure modes is not provided.
- Operational external validity: no production deployment, operator-in-the-loop workload measurement, nor multi-vendor field trials were performed; operator-cost trade-offs remain unquantified at scale.

VII. CONCLUSION

We executed a fully archived preregistered paired experiment comparing `gpt-5-mini` baseline generation and a deterministic verifier-augmented pipeline on 200 synthetic NetOps tasks. Under the executed `shared_initial_candidate` semantics and for the declared model, both arms produced zero verifier-detected unsafe proposals (paired difference = 0.0; exact McNemar $p = 1.0$; 95% CI = [0.0, 0.0]). Because identical per-pair generations were used and because the observed baseline unsafe rate was 0%, the preregistered causal hypothesis was not testable in this run. The scientific contribution is therefore methodological and reproducibility-centred: (1) a complete deterministic preregistered run with archived artifacts that permit exact replay; (2) an artifact-grounded diagnosis of why this execution was non-informative for verifier efficacy; and (3) concrete, archive-supported recommendations (independent per-arm generation budgeting, adversarial calibration, and exploratory ROC estimation) for informative repeats. The archived execution and analysis artifacts cited throughout (`execution/*` and `analysis/*`) permit deterministic replication of the diagnosis and the run outputs and should be consulted prior to attempting repeats or production extrapolation.

DISCLOSURE STATEMENT

Disclosure Statement (CNSM Agentic AI Researcher track): Declared LLM: `gpt-5-mini` (execution recorded in `execution/model_configuration.json`). Orchestration/framework: `hosted_netops_gvr_v1` adapter and deterministic experiment harness (`execution/execution_manifest.json`).

Exact initial master prompt: archived at `provenance/master_prompt.txt` with immutable SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15`. Topic constraints and autonomy boundary: the human Research Director selected the broad topic family (Generative AI and LLMs for NetOps) prior to the autonomy lock; after the lock the autonomous researcher performed literature review, preregistration (`cand-2026-01-prereg-v1`), experiment design, code generation, execution, analysis, manuscript drafting, autonomous peer review, and revisions. Preregistration and execution: the preregistration was frozen before execution; key preregistered elements (estimand, paired design $N=200$, task generator, allowed generation semantics, scoring rules, and stopping rules) are recorded in `cand-2026-01-prereg-v1` and archived in the manuscript bundle. Generation semantics and model-call accounting (executed): `independent_condition_generation = false` (`shared_initial_candidate` semantics) producing a single initial LLM candidate per task shared across arms; `model_calls_used = 200` while the preregistered `maximum_model_calls` allocated for independent per-arm generation = 400 (`execution/execution_manifest.json` and `execution/model_configuration.json`). Validator and scoring: `deterministic_netops_validator_v5` performed deterministic sandboxed validation and encoded outcomes as binary labels (1 = SAFE, 0 = UNSAFE) under `scoring_rule = full_intent_constraint_validation`; archived scoring JSONs (`execution/scoring/*json`) contain validator traces showing `validation_before/after` and violation lists. Analysis executor: `paired_binary_analysis_v1` computed the paired contingency, exact McNemar two-sided test, and paired bootstrap CI (resamples = 10000, seed = 1729); analysis artifacts include `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv`. Deviations and restarts: no post-preregistration design changes were executed; the observed model call count reflects the executed `shared_initial_candidate` semantics. Reproducibility pointers (concise): `execution/raw_results.jsonl`, `execution/task_manifest.jsonl`, `execution/model_configuration.json`, `execution/execution_manifest.json`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, and `analysis/paired_difference_figure.svg` are archived in the supplied bundle and permit deterministic replays. Human editing: no manual human edits to technical content, analyses, or manuscript text occurred after the autonomy lock. Pre-lock development: infrastructure rehearsals occurred prior to the lock but were excluded from final empirical evidence unless reproduced under the post-lock execution.

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